

DataHarmonizer: Mitigating Airborne PCB Volatilization from Estuarine Sediment Using Bacterial Biofilms Attached to Corn-Kernel Biochar Data (Project 5)

Purpose

The ISRP DataHarmonizer is a standardized browser-based spreadsheet editor and validator that can reinforce using standardized columns (AKA fields, variables) and controlled vocabulary for certain values (e.g., NA for missing values). Using the DataHarmonizer can also help ensure your data meets our ISRP tabular data structure guideline and improve data exchange between ISRP research projects and support cores.

Template

The templates were created using datasets from David Ramotowski's PCB biodegradation experiments in Project 5. Each dataset should have its own template.

Procedure

1. Download the zip file (DataHarmonizer_Project5_David_June_18 (zip)) containing the DataHarmonizer application from the following link: {TBD}
2. Extract the zip file and navigate into the extracted folder. Open main.html. The DataHarmonizer will open in your default browser. Click **File** on the top left toolbar, and then click **Select Template**. Now you'll see a pop-up window. Pull the downwards arrow to change the template, and then click **Open**.
3. To import data from local xlsx, xls, tsv and csv files to the DataHarmonizer, click **File** on the top left toolbar, and then click **Open**. You can also copy and paste data from a spreadsheet to the DataHarmonizer.
4. Definitions, examples, and guidance about field headers are available by double clicking on the field headers, or by using the Reference Guide. To access the Reference Guide, click on the **Help** button, and then click **Reference Guide**.

Required fields are organized into subsections.

Required Fields	Field Name	Subsection	Template Name
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sample id	sample_id	Sampling Metadata	David_03_rawPCBmass_PUF David_09_RT-qPCR_data
experiment	experiment	Sampling Metadata	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME David_09_RT-qPCR_data
sample medium	sample_medium	Sampling Metadata	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_05_Dataset_rawBlankmass David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME
units	units	Sampling Metadata	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_05_Dataset_rawBlankmass David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME David_09_RT-qPCR_data
time	time	Sampling Metadata	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME David_09_RT-qPCR_data
group	group	Sampling Metadata	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME David_09_RT-qPCR_data
replicate	replicate	Sampling Metadata	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME David_08_Dataset_NBH_SedimentPCBs
percent biochar	percent_biochar	Sampling Metadata	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME David_09_RT-qPCR_data
blank ID	blank_id	Sampling Metadata	David_05_Dataset_rawBlankmass

location	location	Sampling Metadata	David_08_Dataset_NBH_SedimentPCBs
latitude	latitude	Sampling Metadata	David_08_Dataset_NBH_SedimentPCBs
longitude	longitude	Sampling Metadata	David_09_RT-qPCR_data
experiment replicate	experiment_replicate	Sampling Metadata	David_09_RT-qPCR_data
tPCB	tPCB	PCB Measurements	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_05_Dataset_rawBlankmass David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME David_08_Dataset_NBH_SedimentPCBs
LCPCB	LCPCB	PCB Measurements	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_05_Dataset_rawBlankmass David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME David_08_Dataset_NBH_SedimentPCBs
PCB1-PCB209	PCB1-PCB209	PCB Measurements	David_03_rawPCBmass_PUF David_04_rawPCBmass_SPME David_05_Dataset_rawBlankmass David_06_final_PCBmass_PUF David_07_final_PCBmass_SPME David_08_Dataset_NBH_SedimentPCBs
DNA	DNA	Sampling Metadata	David_09_RT-qPCR_data
cDNA	cDNA	Sampling Metadata	David_09_RT-qPCR_data

All PCB analyte fields are required. If a value is unavailable, enter NA. If the analyte is below the limit of quantification, enter ND. Do not leave cells blank.

Some fields contain a picklist for predefined values. For example, PCB fields contain ND (under detection) and NA (missing value).

- Validate the imported data by clicking on the **Validate** button on the top left toolbar. Missing values and invalid entries in required fields will be highlighted in red. Address any invalid data that was flagged in red in the template.
 - Pale Red = Incorrect data format

- Dark Red = Required data missing

Note: It is possible to export incomplete or invalid data. Make sure to review any errors prior to exporting.

6. Once you've completed validating your data, save data by clicking **File**, and then Save as. Enter the file name and press **Save**. The file will be automatically saved in Downloads. Files saved in the DataHarmonizer are (.xlsx or .csv) format and can be re-opened directly.